Practice Set 3 Paper 2 – Urology

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Practice paper compiled from the question bank.

Q1. Describe procedure of identification, maintenance, distribution and retrieval of kidney in a 25 yr old brain dead pt

Type: LAQ (25 marks) (also answered in 2021-Summer Paper 2, Q2 and 2022-Summer Paper 4, Q3b)

Assumed meaning: the Indian deceased-donor pathway: identification and brain-stem death certification, donor maintenance, allocation and retrieval.

Introduction

- A 25-year-old brain-dead patient is an ideal **standard-criteria DBD (donation after brain death)** donor: circulation continues until cold perfusion, so warm ischaemia is minimal.
- **THOA (Transplantation of Human Organs Act) 1994**, amended 2011, **Rules 2014**, makes brain-stem death legal death.

1. Identification

- **Potential donor:** ventilated patient with catastrophic brain injury (head injury, intracranial bleed, hypoxia). ICU (intensive care unit) informs the **transplant coordinator**; **mandatory request** to the family.
- **Evaluation:** history, creatinine, urine output, ultrasound; ABO and HLA (human leucocyte antigen) typing; serology and NAT (nucleic acid testing) for HIV (human immunodeficiency virus), hepatitis B and C, CMV (cytomegalovirus), syphilis.
- **Contraindications:** active or metastatic cancer, haematological malignancy, past melanoma, untreatable viral infection.

2. Certification of brain-stem death (THOA)

- **Board of four:** doctor in charge of the hospital; independent specialist from an approved panel; neurologist or neurosurgeon (else a nominated surgeon, physician, anaesthetist or intensivist); treating doctor. **None from the transplant team.**
- **Preconditions:** apnoeic coma with a known irreversible cause; exclude sedatives, relaxants, hypothermia, metabolic causes and shock.

Brainstem reflex (all absent)	Method	Cranial nerves
Pupillary	Bright light: pupils fixed, no reaction	II, III
Corneal	Touch cornea: no blink	V, VII
Oculocephalic (doll's eye)	Turn head: eyes move with the head	VIII, III, VI
Oculovestibular (caloric)	Ice-cold water into each ear (drum seen): no eye movement	VIII, III, VI
Motor response in cranial nerve area	Supraorbital pressure: no grimace	V, VII
Gag and cough	Touch pharynx; suction catheter in trachea: no response	IX, X

- **Apnoea test (last):** pre-oxygenate with 100% oxygen; disconnect the ventilator with oxygen insufflation; **no breathing effort at $PaCO_2$ (arterial carbon dioxide tension) ≥60 mmHg**. Abort for hypotension, desaturation or arrhythmia.
- **Two sets of tests at least 6 hours apart**; **Form 10**. Ancillary tests (angiography, perfusion scan) only if examination is impossible.
- **Consent** of near relative; in medico-legal cases inform police.

3. Maintenance (donor management)

- **Why:** herniation causes a **catecholamine storm**, then loss of sympathetic tone (hypotension), **diabetes insipidus (DI)**, hypothermia, DIC (disseminated intravascular coagulation), low thyroid hormone and cortisol, and pulmonary oedema.
- **"Rule of 100s":** systolic pressure >100 mmHg, urine >100 mL/h, PaO_2 (arterial oxygen tension) >100 mmHg, haemoglobin >100 g/L.
- **Measures:** fluids guided by CVP (central venous pressure), no starch colloids (nephrotoxic); **vasopressin** first (pressor and treats DI), then noradrenaline; methylprednisolone 15 mg/kg, thyroid hormone, insulin; lung-protective ventilation; warming; correct sodium, potassium and clotting.

4. Distribution (allocation)

- **NOTTO (National Organ and Tissue Transplant Organisation)**, New Delhi, with **ROTTOs** (regional) and **SOTTOs** (state organisations), keeps registries and waiting lists.
- **Sequence:** hospital and state list → region → national.
- **Kidney criteria:** ABO compatibility, waiting time on dialysis, HLA match, priority for children and previous living donors, **negative crossmatch** (donor spleen or nodes). Many states let the retrieving hospital keep one kidney.
- 2023 policy removed the domicile requirement and upper age limit for registration.

5. Retrieval (multi-organ, DBD)

1. Check certificate, consent, blood group. **Incision:** xiphoid-to-pubis midline with sternotomy.
2. **Exposure:** mobilise right colon and small-bowel mesentery (Cattell–Braasch manoeuvre); expose aorta and IVC (inferior vena cava). **Look for aberrant renal arteries**; keep accessory arteries from the iliacs.
3. **Cold perfusion:** heparin; cannulate the distal aorta; clamp the supracoeliac aorta; **vent** the IVC; flush cold UW (University of Wisconsin) or HTK (histidine–tryptophan–ketoglutarate) solution; crushed ice; open Gerota's fascia laterally.
4. **Heart, lungs, liver and pancreas out first.**
5. **Kidneys en bloc:** lift with perinephric fat off the posterior wall; keep periureteric tissue; divide ureters low, then aorta and IVC (upper limit set by the liver team); divide prevertebral tissue.
6. **Extras:** iliac vessels; spleen and nodes for crossmatch.

6. Back-bench preparation and preservation

- Divide the aorta between renal ostia (**Carrel patches**); keep IVC with the right kidney to lengthen its vein; ligate lymphatics; triple-bag on ice.
- Cooling to 4 °C slows metabolism. **Hypothermic machine perfusion** reduces DGF (delayed graft function), not long-term graft loss. Cold ischaemia >24 hours raises thrombosis risk.

Diagram to draw: aorta and IVC with distal aortic cannula, supracoeliac clamp, IVC vent and en bloc kidneys.

> **Key points to remember:**
>
> - Board of four, none from the transplant team; two tests ≥6 h apart; Form 10.
> - Apnoea: no breathing at $PaCO_2$ ≥60 mmHg.
> - Rule of 100s; vasopressin for hypotension and DI.
> - Distal aortic cannula, supracoeliac clamp, IVC vent; kidneys out last.

Q2. Management of bone mets in prostatic CA

Type: LAQ (25 marks) (mHSPC with retention and bone pain also answered in Practice Set 1 Paper 2, Q1)

Introduction

- Bone is the commonest metastatic site; visceral metastasis occurs in <20%.
- Aims: control the cancer, relieve pain, **prevent SREs (skeletal-related events)** and treat emergencies (cord compression, fracture).

Pathophysiology

- Tumour cells lodge in axial marrow, adhering through integrins and collagen receptors.
- Metastases are mainly **osteoblastic (sclerotic)**, but resorption also rises. Tumour cytokines induce **RANKL (receptor activator of nuclear factor-κB ligand)**, which activates osteoclasts; released growth factors feed the tumour (**vicious cycle**). ADT (androgen deprivation therapy) adds osteoporosis through loss of oestrogen.
- **SREs:** pathological fracture, spinal cord compression, need for radiotherapy or surgery to bone. Marrow replacement causes anaemia. Hypercalcaemia is rare and suggests neuroendocrine disease.

Evaluation

- **Clinical:** pain, neurological signs, performance status. **Bloods:** PSA (prostate-specific antigen), ALP (alkaline phosphatase), calcium, creatinine, haemoglobin, testosterone.
- **Imaging:** bone scan and CT (computed tomography) define volume. PSMA (prostate-specific membrane antigen) PET is more sensitive, but trials used conventional imaging. **Plain X-ray or CT of painful weight-bearing sites** (lytic lesion, impending fracture). **Whole-spine MRI (magnetic resonance imaging)** for back pain with neurological signs.
- **Volume (CHAARTED):** high volume = ≥4 bone metastases with ≥1 outside the axial skeleton, or visceral metastasis.

1. Systemic treatment of the cancer

Hormone-sensitive (mHSPC):

- **ADT** (orchidectomy, LHRH [luteinising hormone-releasing hormone] agonist or antagonist). **Flare:** an agonist raises testosterone for 10–20 days and can worsen pain or cord compression. Cover it with an antiandrogen for 21–28 days, or use an antagonist (degarelix) or orchidectomy.
- **Intensify:** add abiraterone + prednisone (LATITUDE), enzalutamide (ENZAMET, ARCHES) or apalutamide (TITAN), or a **triplet** with docetaxel (PEACE-1, ARASENS). **Prostate radiotherapy** for low-volume synchronous disease (STAMPEDE).

Castration-resistant (mCRPC):

- ARPIs (androgen receptor pathway inhibitors), docetaxel then cabazitaxel, PARP (poly-ADP-ribose polymerase) inhibitors for HRR (homologous recombination repair) mutations.
- **Radium-223** (alpha emitter, path <100 μm, spares marrow): for **symptomatic bone-only** disease; 50 kBq/kg every 4 weeks × 6. **ALSYMPCA:** overall survival 14.9 vs 11.3 months; delayed SREs. **Never with abiraterone** (more fractures and deaths).
- **^{177}Lu-PSMA-617** (beta emitter, PSMA-avid disease at any site): after ARPI and taxane; **VISION:** survival 15.3 vs 11.3 months.

2. Bone-targeted agents

Feature	Zoledronic acid	Denosumab
Mechanism	Bisphosphonate: inhibits osteoclasts	Antibody against RANKL
SRE dose (mCRPC)	4 mg IV every 4 weeks, adjust for creatinine clearance	120 mg SC every 4 weeks, no renal adjustment
Evidence	Fewer SREs than placebo	Time to first SRE 20.7 vs 17.1 months vs zoledronate
Toxicity	Fever, myalgia, renal impairment, hypocalcaemia, ONJ	Hypocalcaemia, hypophosphataemia, ONJ 2–4%

- **No survival benefit**; they protect the skeleton only.
- **In mHSPC, SRE doses are not indicated** (no benefit, e.g. CALGB 90202). Use osteoporosis doses for ADT bone loss: denosumab 60 mg every 6 months or zoledronic acid yearly, by DEXA (dual-energy X-ray absorptiometry) and FRAX.
- **ONJ (osteonecrosis of the jaw) prevention:** dental check before starting; calcium 1000 mg and vitamin D 800 IU daily.
- Give a bone agent with radium-223 or any ARPI.

3. Pain

- **Analgesic ladder** (WHO [World Health Organization]); neuropathic pain needs gabapentinoids or tricyclics; steroids for diffuse pain.
- **Focal pain: EBRT (external beam radiotherapy) single 8 Gy fraction.** Pain relief equals 30 Gy in 10 fractions (RTOG 9714); retreatment is commoner.
- **Diffuse pain:** wide-field RT; radiopharmaceuticals (strontium-89, samarium-153 relieve pain only; radium-223 adds survival).
- **SBRT (stereotactic body RT)** for oligometastases (STOMP, ORIOLE) and spinal re-irradiation.

4. Emergencies

Spinal cord compression (every focal back pain is compression until proven otherwise):

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Back pain ± weakness, sensory level, sphincter change
 → dexamethasone 10 mg IV load, then 4–10 mg 6–12-hourly
 → urgent whole-spine MRI
 → decompressive surgery + RT (unstable spine, progression on RT, prior RT)
   or RT alone
 → orchidectomy or antagonist if hormone-naive (never agonist alone)
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- Surgery plus RT beat RT alone for walking ability (**Patchell, 2005**).

Pathological or impending fracture: orthopaedic fixation then RT. **Mirels score** ≥9 (site, pain, lytic type, size) favours prophylactic fixation.

Follow-up

- PSA, ALP, calcium, renal function every 3–6 months; new scan if symptoms or PSA rise. PCWG3 (Prostate Cancer Working Group 3): bone scan progression needs ≥2 new lesions, confirmed later (early "flare" misleads).
- **Also know:** stent or nephrostomy for obstruction; marrow-failure anaemia; hospice care.

Diagram to draw: vicious cycle: tumour → RANKL → osteoclast → growth factors → tumour, with the sites blocked by denosumab and zoledronate.

Key points to remember:

- Avoid flare: antiandrogen cover, antagonist or orchidectomy.
- Single 8 Gy fraction for focal pain; dexamethasone and MRI for cord compression.
- Denosumab 120 mg or zoledronate 4 mg 4-weekly prevent SREs in CRPC, not in HSPC.
- Radium-223 (ALSYMPCA) for symptomatic bone-only mCRPC; never with abiraterone.

Q3a. BCG failure

Type: SAQ (10 marks) (BCG therapy also answered in 2025-Winter Paper 1, Q3f)

Introduction

- BCG (bacille Calmette–Guérin) is standard for high-risk NMIBC (non-muscle-invasive bladder cancer). "Failure" was poorly defined; consensus categories now guide treatment.

Definitions

Category	Definition	Implication
Adequate BCG	≥5 of 6 induction + ≥2 of 3 maintenance, or ≥2 doses of a second induction; for T1, ≥5 of 6 induction alone	Threshold for "unresponsive"
BCG-unresponsive	Persistent or recurrent CIS (carcinoma in situ) or HG (high-grade) Ta/T1 despite adequate BCG; or HG relapse within **6 months (Ta/T1) or 12 months (CIS)** of last dose; **HG T1 at first 3-month check**	More BCG futile; cystectomy
BCG-resistant	CIS or HG Ta at 3 months after induction	Reinduction: 30–50% respond
Delayed relapse	HG relapse beyond these windows	Rechallenge may work
BCG-intolerant	Toxicity prevented adequate BCG	Salvage chemotherapy

- **Why BCG fails:** inadequate dosing; PD-L1 (programmed death-ligand 1) upregulation on tumour cells.

Evaluation

- Cystoscopy, cytology, **re-TURBT (transurethral resection of bladder tumour)** and biopsies to exclude invasion.
- **Extravesical disease:** prostatic urethral biopsy (24–30%) and CT urography (upper tract 0.8–25%).

Management

1. **Timely radical cystectomy** (EAU [European Association of Urology], AUA [American Urological Association]): HG T1 at 3 months progresses in 82%; CIS failure in about 50%.
2. **Bladder sparing** (unfit or refusing):

Option	Key result (CIS complete response, CR)
Pembrolizumab IV (KEYNOTE-057)	41% at 3 months, 19% at 12 months; grade 3–4 toxicity 13%
Nadofaragene firadenovec (adenovirus carrying IFN-α2b [interferon alpha-2b] gene) every 3 months	53% at 3 months, 24% at 12 months
Gemcitabine–docetaxel sequential	60% at 1 year; 84% avoid cystectomy (retrospective)
N-803 + BCG (IL-15 [interleukin-15] superagonist)	45% at 12 months

- **Also know:** BCG-refractory (older term); gemcitabine, gemcitabine–mitomycin, valrubicin, chemohyperthermia, EMDA (electromotive drug administration), chemoradiation; TAR-200; cretostimogene.

Diagram to draw: flowchart: HG recurrence after BCG → adequate? → unresponsive → cystectomy or bladder sparing.

Key points to remember:

- Adequate BCG = 5/6 induction + 2/3 maintenance.
- Unresponsive: HG within 6 months (Ta/T1) or 12 months (CIS); HG T1 at 3 months.
- Timely cystectomy is standard; salvage CR rates halve by 12 months.

Q3b. Describe various pt positions in PCNL

Type: SAQ (10 marks) (also answered in 2024-Winter Paper 1, Q3f)

Introduction

- PCNL (percutaneous nephrolithotomy) position decides calyceal access, working space, airway safety and simultaneous retrograde access.

Positions

Position	Set-up	Advantages	Disadvantages
Prone (Goodwin, 1955)	Lithotomy for ureteric catheter, then turned prone; horizontal bolsters at nipple line and ASIS (anterior superior iliac spine) bring the kidney down more than vertical ones	**Posterior calyx and upper pole** access; large working area; familiar	Turning; airway hard to reach; pressure injuries; retrorenal colon commoner prone
Prone split-leg / prone-flexed	Legs apart for flexible cystoscopy; hip flexion opens the costovertebral space	ECIRS (endoscopic combined intrarenal surgery) prone	Needs split-leg table; prone cystoscopy learning curve
Supine (Valdivia-Uría, 1987)	Ipsilateral flank at table edge on a lumbar bolster; arm across chest	Airway retained; no turning; **gravity drains fragments**, low pelvic pressure	Less working area; anterior calyces; mobile kidney; harder upper pole
Galdakao-modified supine Valdivia (Scoffone)	Supine flank tilt with ipsilateral leg extended and other leg in lithotomy	**Simultaneous retrograde access (ECIRS)** in one position	Longer, more horizontal tract
Bart's flank-free modified supine (Bach)	Bolster under the thorax, flank free	More flank space	Fewer data
Lateral decubitus (flank)	Kidney-up flank position	Obese, kyphoscoliosis, poor respiratory reserve	Unfamiliar fluoroscopic orientation

Positioning safety (prone)

- Bolster shins so toes float; pad knees; arms ≤90° to the torso (brachial plexus); axillary and pectoral rolls; padded face and eyes.
- **Prone-specific complications:** neuropraxia, ocular injury, endotracheal tube displacement.

Choosing a position

- **Prone:** standard anatomy, upper pole or multiple posterior tracts.
- **Supine variants:** obesity, cardiopulmonary disease, ECIRS, ileal conduit or continent channel, combined ureteric stone.
- **Spinal deformity:** individualise; pad every pressure point.
- **Evidence:** similar stone-free and complication rates; choice is mainly surgeon preference.

Diagram to draw: axial CT sketch showing prone (vertical) and supine (horizontal) tracts.

Key points to remember:

- Prone: upper pole and posterior calyx access, large working area.
- Galdakao-modified supine: one position for ECIRS, airway safe, gravity drainage.
- Outcomes equal; horizontal bolsters drop the kidney more.

Q3c. Grade 4 Renal trauma.

Type: SAQ (10 marks) (renal trauma also answered in 2017-Summer Paper 2, Q2)

Definition: AAST (American Association for the Surgery of Trauma) 2018, grade IV

- Laceration into the **collecting system with urinary extravasation**.
- **Renal pelvis laceration and/or complete UPJ (ureteropelvic junction) disruption.**
- **Segmental artery or vein injury** (pseudoaneurysm, AVF [arteriovenous fistula]).
- **Active bleeding beyond Gerota's fascia.**
- **Segmental or complete infarction** from thrombosis without active bleeding.
- Change from 1989: main-vessel injury moved to grade V; segmental vascular and UPJ injuries added to grade IV.

Diagnosis

- Haematuria may be **absent** in grade IV, pedicle and UPJ injuries; mechanism matters.
- **Contrast CT (computed tomography) with delayed phase** (5–10 minutes): arterial phase for bleeding, delayed phase for urine leak.
- **High-risk signs:** haematoma rim >3.5–4 cm, **vascular contrast blush**, medial laceration (≥2 predict intervention). Medial extravasation without distal ureteric contrast = UPJ avulsion.

Management

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Unstable despite resuscitation → angioembolisation (AE) if team ready, else laparotomy
Stable → non-operative: bed rest, serial haemoglobin every 4-6 h
   Active blush / pseudoaneurysm / AVF → superselective AE
   Urinary extravasation → observe (90% heal)
        stent ± nephrostomy if urinoma enlarges, fever, obstruction,
        or UPJ / pelvis injury
```

- **Non-operative success** about 85% for grade IV–V even in penetrating series; most failures need only AE, stent or nephrostomy.
- **Failure:** instability; falling haemoglobin despite 2–3 units.
- **Repeat CT at ~48 hours** for deep lacerations or fever, pain, falling haematocrit (routine reimaging debated).
- **Exploration:** pulsatile expanding haematoma, instability, UPJ avulsion. Control vessels before opening Gerota's fascia; renorrhaphy or partial nephrectomy; omental cover; drain. Devitalised parenchyma >25% lowers the threshold.

Complications

- Delayed bleeding (within 3 weeks; embolise); urinoma, abscess; **hypertension** (Page kidney, segmental ischaemia, AVF); function loss.

Follow-up

- Blood pressure, urinalysis, creatinine; DMSA (dimercaptosuccinic acid) scan for salvaged kidneys (EAU, European Association of Urology).

Diagram to draw: kidney with laceration into a calyx, urinoma and segmental pseudoaneurysm.

Key points to remember:

- Grade IV = collecting system, UPJ, segmental vessel, bleeding beyond Gerota's, infarction.
- Delayed CT phase is essential; blush means AE.
- Most stable grade IV heal non-operatively; extravasation resolves in up to 90%.

Q3d. Algorithm for management of an incidentaloma

Type: SAQ (10 marks) (also answered in 2022-Summer Paper 4, Q3f)

Assumed meaning: adrenal incidentaloma.

Definition

- Adrenal mass **≥1 cm** found on imaging done for another reason. Two questions: **Is it malignant (imaging)? Is it functioning (biochemistry)?**
- Causes: non-functioning adenoma (commonest); cortisol-secreting adenoma 5.3%; phaeochromocytoma 5.1%; aldosteronoma 1%; ACC (adrenocortical carcinoma) 4.7%; metastasis 2.5%; myelolipoma, cyst.

Algorithm

```
Adrenal mass ≥1 cm
 ├─ IMAGING: unenhanced CT (computed tomography)
 │    ≤10 HU (Hounsfield units), homogeneous → benign adenoma
 │    >10 HU → washout CT (15 min) or chemical-shift MRI
 │         APW >60% or RPW >40% → adenoma
 │         indeterminate → MDT: repeat CT 6-12 months, FDG-PET or surgery
 │    Macroscopic fat → myelolipoma
 ├─ HORMONES (all patients)
 │    1-mg overnight DST (dexamethasone suppression test)
 │    plasma or urine metanephrines
 │    ARR (aldosterone-renin ratio) if hypertension or low potassium
 │    sex steroids only if ACC suspected
 └─ DECISION
      functioning, suspicious, >4 cm, or growing → adrenalectomy
      benign, non-functioning → discharge
```

- **APW** (absolute percentage washout) = (enhanced − delayed) ÷ (enhanced − unenhanced) × 100; **RPW** (relative) = (enhanced − delayed) ÷ enhanced × 100.

Interpreting the tests

- **DST:** cortisol ≤1.8 μg/dL (50 nmol/L) excludes; 1.8–5.0 = possible autonomous secretion (**MACS**, mild autonomous cortisol secretion); >5.0 μg/dL autonomous.
- **Size and malignancy:** <4 cm 0.1%; 4–6 cm 2.4%; >6 cm 19.5%. Benign imaging overrides size.
- **Never biopsy or operate before excluding phaeochromocytoma.** Biopsy only for suspected metastasis in known cancer.

Management

- **Surgery:** cortisol excess, phaeochromocytoma (after alpha-blockade), aldosteronoma (after AVS, adrenal vein sampling), malignant features, >4 cm in fit patients, significant growth. Laparoscopic for benign; open for suspected ACC.
- **MACS:** screen for hypertension, diabetes, osteoporosis; surgery individualised.
- **Follow-up (ESE–ENSAT 2023, European Society of Endocrinology–European Network for the Study of Adrenal Tumors):** homogeneous ≤10 HU, non-functioning → no further imaging. Indeterminate non-resected → re-image at 6–12 months; growth >20% and ≥5 mm is concerning.
- **Bilateral:** metastases, lymphoma, tuberculosis, haemorrhage, CAH (congenital adrenal hyperplasia; 17-hydroxyprogesterone); exclude adrenal insufficiency.

Diagram to draw: the algorithm.

Key points to remember:

- ≤10 HU = lipid-rich adenoma; APW >60%, RPW >40%.
- 1-mg DST >1.8 µg/dL = MACS.
- Exclude phaeochromocytoma before biopsy or surgery.
- Surgery: functioning, suspicious, >4 cm or growing.

Q3e. Metabolic abnormalities in CKD

Type: SAQ (10 marks)

Definition

- **CKD (chronic kidney disease), KDIGO (Kidney Disease: Improving Global Outcomes):** abnormal kidney structure or function for **≥3 months**. Staged by GFR (glomerular filtration rate) G1–G5 (G3a 45–59, G3b 30–44, G4 15–29, G5 <15 mL/min/1.73 m^2) and albuminuria A1–A3.
- Each abnormality is a lost kidney function: excretion, acid–base balance, hormone production.

Abnormalities

Abnormality	Mechanism	Management
Sodium and water	Fixed urine concentration; sodium retention → overload, hypertension	Salt restriction; diuretics (chlorthalidone works even in advanced CKD)
Hyperkalaemia	Less distal sodium delivery; RAAS (renin–angiotensin–aldosterone system) blockers lower aldosterone	Potassium binders and diuretics, so RAAS blockers can continue
Metabolic acidosis	Less ammonium excretion; retained sulphate and phosphate (anion gap rises late); bicarbonate loss	Sodium bicarbonate or citrate to keep bicarbonate **≥22 mEq/L**; slows progression, protects bone and muscle
CKD-MBD (mineral and bone disorder)	Phosphate retention → FGF-23 (fibroblast growth factor 23) rises → less **calcitriol** → hypocalcaemia → **secondary hyperparathyroidism**	Phosphate restriction, binders with meals, active vitamin D; calcimimetic; parathyroidectomy if tertiary
Anaemia	Erythropoietin falls, mainly below GFR 30	Iron first, then erythropoietin; Hb (haemoglobin) target 10–11.5 g/dL (harm above 13)
Hyperuricaemia	Reduced excretion	Treat gout; asymptomatic lowering unproven (CKD-FIX)
Lipids	High triglycerides, low HDL (high-density lipoprotein)	Statin ± ezetimibe if ≥50 years, not on dialysis (SHARP)
Glucose	Insulin resistance, but slower insulin clearance → hypoglycaemia	Lower insulin doses

Bone disease (renal osteodystrophy)

- **High turnover:** osteitis fibrosa (PTH [parathyroid hormone] excess).
- **Low turnover:** adynamic bone (oversuppressed PTH), osteomalacia.
- **Tertiary hyperparathyroidism:** autonomous gland, hypercalcaemia; persists after transplant.
- **Vascular calcification** adds cardiovascular death.

Other endocrine and uraemic effects

- **Also know:** hypogonadism, hyperprolactinaemia, erectile dysfunction (better after transplant); growth failure in children; protein-energy wasting; dialysis amyloid; uraemic platelet dysfunction (desmopressin).

- **Urological relevance:** dose drugs by GFR; avoid gadolinium if eGFR (estimated GFR) <30; each AKI (acute kidney injury) speeds progression.

Diagram to draw: CKD-MBD cascade: phosphate ↑ → FGF-23 ↑ → calcitriol ↓ → calcium ↓ → PTH ↑ → bone disease.

> **Key points to remember:**
> - FGF-23 rises first; secondary then tertiary hyperparathyroidism.
> - Bicarbonate ≥22 mEq/L; Hb 10–11.5 g/dL.
> - Hyperkalaemia: binders rather than stopping RAAS blockade.

Q3f. Gleason's score and its significance

Type: SAQ (10 marks)

Introduction

- **Gleason grading** (Donald Gleason, late 1960s) grades prostate adenocarcinoma by **glandular architecture at low magnification**, not cytology. Patterns run from 1 (most differentiated) to 5. Modified by ISUP (International Society of Urological Pathology) in **2005** and **2014**.

Patterns (modern)

Pattern	Architecture
1–2	Not used on biopsy now
3	Discrete, well-formed glands of any size
4	**Cribriform** (including glomeruloid), poorly formed or fused glands
5	Sheets, cords, single cells, solid nests, **comedonecrosis**

Scoring rules

- **Radical prostatectomy:** primary (commonest) + secondary pattern.
- **Biopsy (2005):** commonest + **highest** pattern, however small. Single pattern is doubled (3+3).
- A tertiary pattern 5 under 5% at prostatectomy is reported as a comment; it adds risk but does not change the grade group.
- Report **% pattern 4** in score 7; report cribriform and intraductal carcinoma (both adverse).

ISUP Grade Groups (2014; WHO [World Health Organization] 2016)

Grade group	Gleason score	5-year BCR (biochemical recurrence)-free after prostatectomy
1	3+3=6	96%
2	3+4=7	88%
3	4+3=7	63%
4	8 (4+4, 3+5, 5+3)	48%
5	9–10	26%

- Grade groups separate 3+4 from 4+3 and 8 from 9–10. Starting at "1" avoids patients misreading the lowest score, 6, as moderate cancer.

Significance

- **Most powerful biopsy predictor** of pathological stage, EPE (extraprostatic extension), nodal spread (0% in GG1, 4.3% in GG3), recurrence and death.
- **Risk groups** (D'Amico, NCCN [National Comprehensive Cancer Network]) combine grade, PSA (prostate-specific antigen) and stage.
- **Treatment choice:** GG1 → active surveillance; selected GG2 with little pattern 4 → surveillance; GG ≥4 → high risk, longer ADT (androgen deprivation therapy) with radiotherapy; also selects for brachytherapy and focal therapy.
- **Limitations:** biopsy undergrades about one-third of GG1 (sampling error); inter-observer variation; after ADT or radiotherapy, grading is unreliable.
- **Also know:** pure GG1 does not metastasise, so "is it cancer?" is debated.

Diagram to draw: Gleason's pattern diagram 3 → 4 → 5.

> **Key points to remember:**
>
> - Biopsy score = commonest + highest pattern.
> - 3+4 (GG2) and 4+3 (GG3) differ: BCR-free 88% vs 63%.
> - GG4 = score 8; GG5 = 9–10.

Rapid Revision Sheet – Practice Set 3 Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe procedure of identification, maintenance, distribution and retrieval of kidney in a 25 yr old brain dead pt (LAQ)

- Board of four, none from the transplant team; two tests ≥6 h apart; Form 10.
- Apnoea: no breathing at $PaCO_2$ ≥60 mmHg.
- Rule of 100s; vasopressin for hypotension and DI.
- Distal aortic cannula, supracoeliac clamp, IVC vent; kidneys out last.

Q2. Management of bone mets in prostatic CA (LAQ)

- Avoid flare: antiandrogen cover, antagonist or orchidectomy.
- Single 8 Gy fraction for focal pain; dexamethasone and MRI for cord compression.
- Denosumab 120 mg or zoledronate 4 mg 4-weekly prevent SREs in CRPC, not in HSPC.
- Radium-223 (ALSYMPCA) for symptomatic bone-only mCRPC; never with abiraterone.

Q3a. BCG failure (SAQ)

- Adequate BCG = 5/6 induction + 2/3 maintenance.
- Unresponsive: HG within 6 months (Ta/T1) or 12 months (CIS); HG T1 at 3 months.
- Timely cystectomy is standard; salvage CR rates halve by 12 months.

Q3b. Describe various pt positions in PCNL (SAQ)

- Prone: upper pole and posterior calyx access, large working area.
- Galdakao-modified supine: one position for ECIRS, airway safe, gravity drainage.
- Outcomes equal; horizontal bolsters drop the kidney more.

Q3c. Grade 4 Renal trauma. (SAQ)

- Grade IV = collecting system, UPJ, segmental vessel, bleeding beyond Gerota's, infarction.
- Delayed CT phase is essential; blush means AE.
- Most stable grade IV heal non-operatively; extravasation resolves in up to 90%.

Q3d. Algorithm for management of an incidentaloma (SAQ)

- ≤10 HU = lipid-rich adenoma; APW >60%, RPW >40%.
- 1-mg DST >1.8 µg/dL = MACS.
- Exclude phaeochromocytoma before biopsy or surgery.
- Surgery: functioning, suspicious, >4 cm or growing.

Q3e. Metabolic abnormalities in CKD (SAQ)

- FGF-23 rises first; secondary then tertiary hyperparathyroidism.
- Bicarbonate ≥22 mEq/L; Hb 10–11.5 g/dL.
- Hyperkalaemia: binders rather than stopping RAAS blockade.

Q3f. Gleason's score and its significance (SAQ)

- Biopsy score = commonest + highest pattern.
- 3+4 (GG2) and 4+3 (GG3) differ: BCR-free 88% vs 63%.
- GG4 = score 8; GG5 = 9–10.